STICS crop model general presentation

1. Introduction : History, purpose, structure and types of application.
2. Description of soil and formalization of energy, water, Nitrogen balance
3. Formalizations of development and growth of crops
4. Need in inputs and care for options
5. Principles for calibration and test

Need in inputs and care for options

N Beaudoin; S Andrianarisoa, D Ripoche, on benhalf of the EPS

ALIMENTATION
AGRICULTURE
ENVIRONNEMENT

INRA

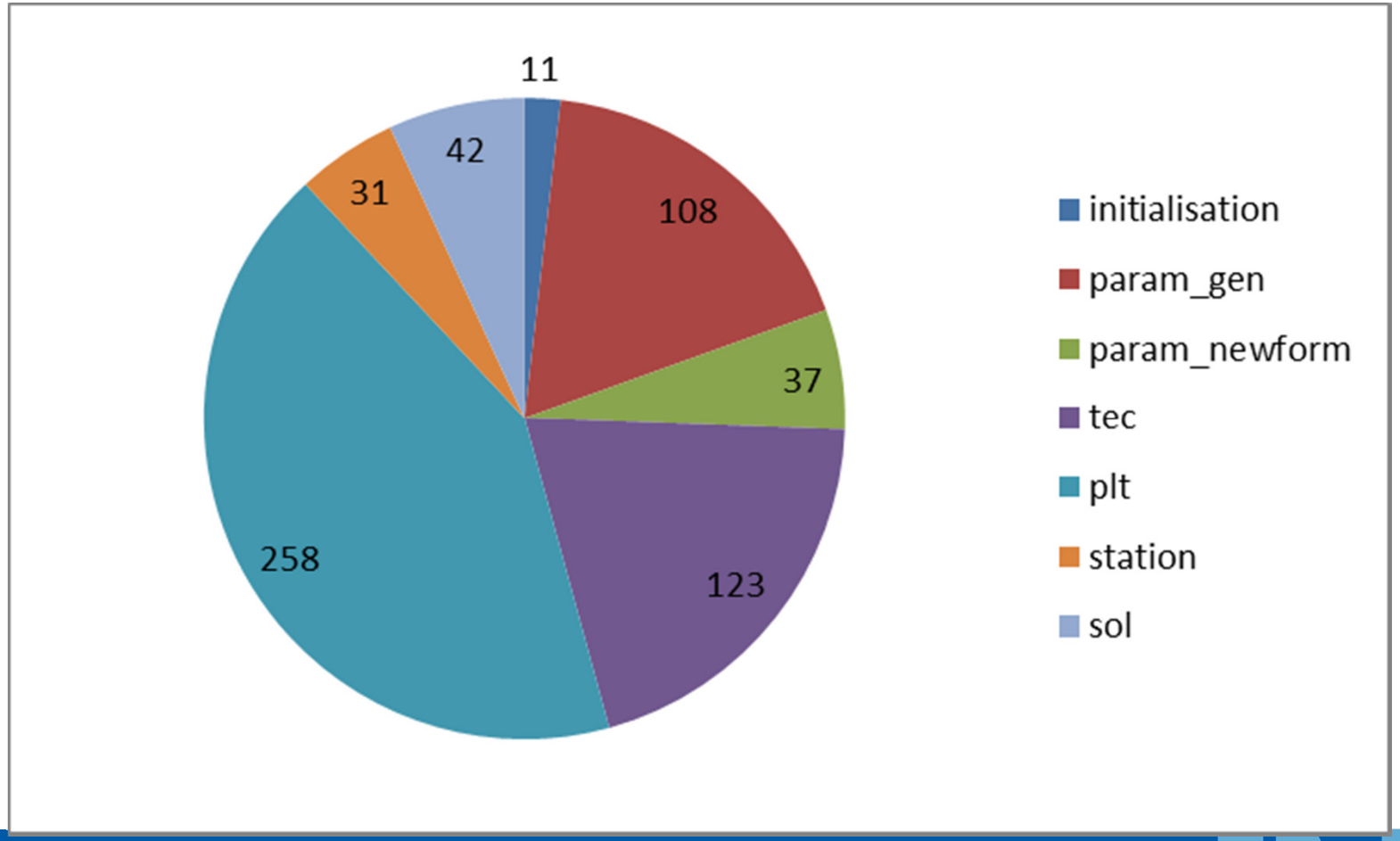
about model inputs

(plt and gen excepted)

STICS

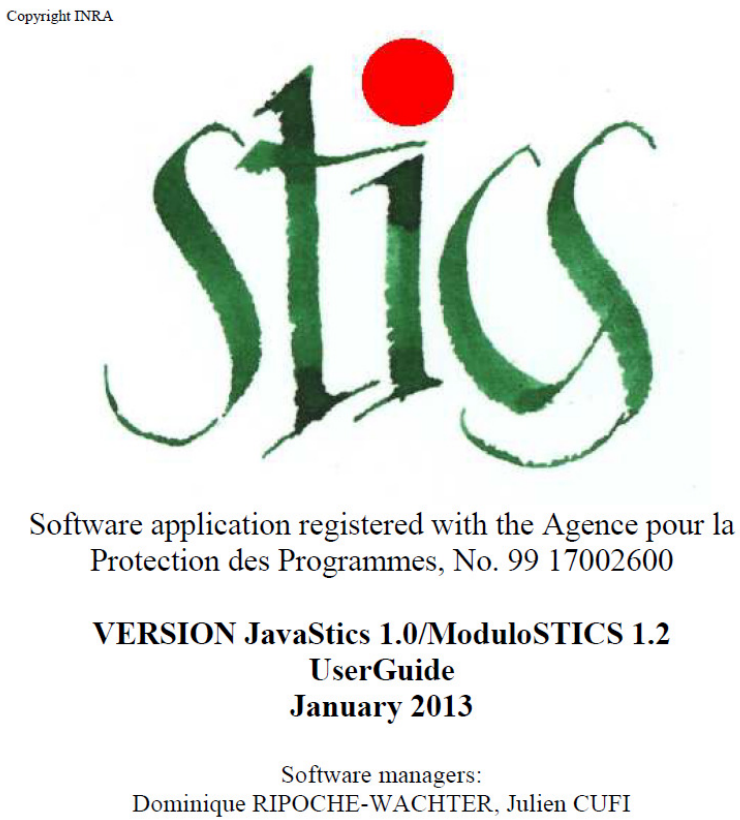
total = 610

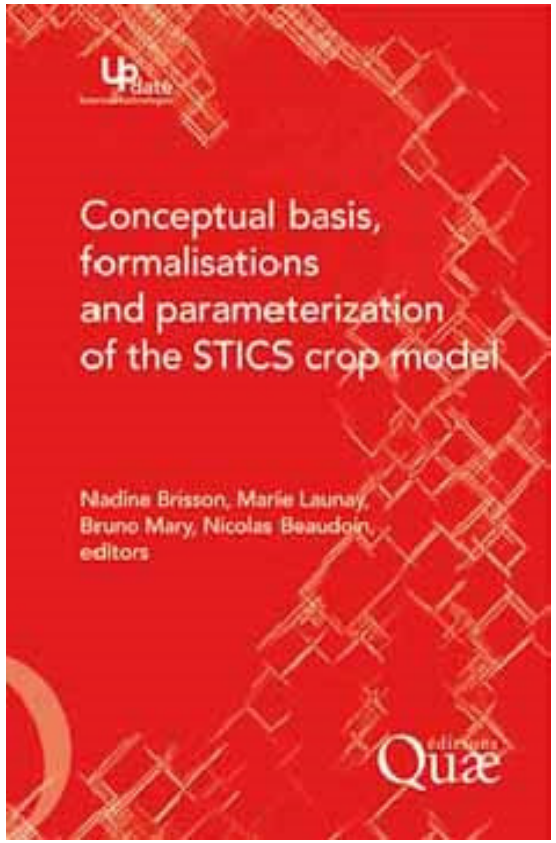
With codeoptions and all options included



How to feed STICS?

1) See hanbooks





2) See Examples within STICS V8modulo

3) Gather database, report or publication about your site.

4) Use new forum STICS

warnings

Initialisations

- Initialisation of soil water content and mineral N = without peebles whereas .obs contains values incluidng them.
- Choosing initial situation as bare soil, i.e. before sowing permits to take 0 for the initial values concerning crop. The latter can carry uncertainty about the initial state.

warnings

Climate

- Distance of climate station : sensitivity depending on the variable (Kovack et al, 1995; Pattey et al., 2012);
- Altitude of the simulated field vs Temperature, wind…
- Latitude need attention because of sun course simulation strongly impacts on radiation and photoperiod.
- PET as input: Penman <> Penman Monteith

Example about climate

Impact of the distance from meterological station:

Taking into account the distance field to the weather station, we found a relationship between increased yields predicted dispersion and distance fields to the weather station. Failure to take into account the spatial distribution of precipitation seems to be the culprit. There should be a distribution grid finer climate data (of the order of 5 x 5 km2).

Pattey et al., 2012; PERFORMANCE OF PREDICTING THE PRODUCTIVITY STICS A GRAIN OF CORN IN RESPONSE TO CLIMATE CHANGE IN EASTERN CANADA. 1) RESET USING THE EXTRACT FROM LANDSAT LAI FOR PREDICTING STPATIALIZED YIELD

Confirming results with CERES (Kovács G.J., 1995) Neméth T., Ritchie J.T., 1995. Testing simulation models for the assessment of crop production and nitrate leaching in Hungary. Agricultural Systems 49: 385-397

warnings

Soil

- Soil Clay = after decarbonatation
- Profhum while no tilled soil => if recent, last plowing depth; if not, highly rooting zone or more Corg concentrated zone.
- Obstarac = maximal rooting depth due to soil (>< crop)
- Bulk density of fine ground if pebbles

warnings

Tec

- Initial value of crop residues: don't miss
- Burying crop residues :
- Profres = layer containing residues upper proftrav.

profres role varying in vs 8 vs 6.9 (in the latter N immobilisation is overestimated if the 'protrav-profres' thickness is too high (Beaudoin et al., 2008)).

- Phenological stage : to force or not? Depends of target; always collect the data, yet. For local diagnosis, forcing emergence date increases model accuracy.

Caring for model options :

- Module activation
- Formalism choice
- Input forcing
- Output choice

GEN Options (general)

15 options level 1

ModuleName	flag	Formalism	Parameter Name
Simulation options			
Nitrogen stress activation	y/n		codeinnact
Water stress activation	y/n		codeh2oact
Optimum mineralisation in bare soil	y/n		codeminopt
Smoothing of initial profiles	y/n		iniprofil
Depth for mineral N and water stocks calculation		profmes/profsol	codeprofmes
Climatic series		reset/succession	codeinitprec
Biomass and yield conservation after harvest	y/n		codemsfinal
Take account of mulch	y/n		codeactimulch
Codefrmur		all fruits (including ripe ones) / growing fruits only	codefrmur
Hourly microclimate	y/n		codemicheur
Scientific writing in st2 and report	y/n		codeoutscien
Separator spaces in report	y/n		codeseprapport
Activation of model sensitivity analysis	y/n		codesensibilite
Water absorption and nitrogen content of the plant			
Nitrogen fixation by legumes		critical nitrogen /nodule activity	codesymbiose
mineral nitrogen inhibition (Nitrogen fixation by legumes = nodule activity)		noeffect / nitrogen amount /nitrogen concentration	codefxn
Soil hydrology and compaction			
calculation of groundwater if drainage		average / interdrain localisation	codhnappe

STA Options (climate station)

5 options level 1

ModuleName	flag	formalism	Parameter Name
Climate			
calculation of potential evapotranspiration		Penman-reading / Penman-calculate / Shutwall-and-Wallace / Prestley-Taylor	codeetp
climate change	y/n		codeclichange
climate in altitude	y/n		codaltitude
option.adret.or.ubac (climate in altitude = y)		adret(south) / ubac(north)	codadret
Microclimate			
calculation of crop temperature		empirical relation / energy balance	codecaltemp
calculation of net radiation		Brunt / Brutsaert	codernet

USM Options

ModuleName	activation	formalism	Parameter Name
optimisation or forcing	Javastics/param.sti		codeoptim
LAI forcing	LAI file		codesimul
rotations	run successive USM		codesuite

SOIL Options

7 options level 1

ModuleName	flag	formalism	Parameter Name
pebbles	y/n		codecailloux
macroporosity	y/n		codemacropor
cracks (case of swelling clay soils)	y/n		codefente
artificial drainage	y/n		codrainage
capillary rise	y/n		coderemontcap
nitrification	y/n		codenitrif
denitrification	y/n		codedenit

PLT Options (1) (plante)

10 options level 1

ModuleName	flag	Formalism	Parameter Name
plant name and clade			
plant name and clade			codeplante
monocot or dicot		monocotyledon / dicotyledon	codemonocot
Phasic development			
driving temperature		air temperature / temperature within canopy	codetemp
time.scale (codetemp = air temperature)		daily.scale / hourly.scale	codegdh
photoperiodic plant	y/n		codephot
delay effect of stress	y/n		coderetflo
cold requirements		no / vernalisation (herbaceous) / dormancy (woody)	codebfroid
dormancy calculation (codebfroid = dormancy (woody))		forcing / Richardson / Bidabe	codedormance
post dormancy calculation (codebfroid = dormancy (woody))		daily temperatures /hourly temperatures	codegdhdeb
Emergence and starting			
annual or perennial		annual / perennial	codeperenne
germination or latency (codeperenne=annual)	y/n		codegermin
plantlet growth (codeperenne=annual)		hypocotyle growth / planting	codehypo
leaf dynamics		LAI / ground cover	codelaitr
LAI calculation option (codelaitr = LAI)		direct LAInet / LAInet=LAIbrut-senes	codlainet
effect of photoperiod on senescence (codelaitr = LAI)	y/n		codestrphot
radiation interception			
radiation interception		Beer's law / radiation transfers	codetransrad

PLT Options (2)

14 options level 1

ModuleName	flag	Formalism	Parameter Name
yield formation			
growing dynamics		determinate growing plant / indeterminate growing plant	codeindetermin
unit Harvest Index (codeindetermin = determinate growing plant)		days / degrees days	codeir
number of inflorescences (codeindetermin = indeterminate growing plant)		prescribed / trophic status function	codcalinflo
thermal stress on filling	y/n		codetremp
roots			
driving temperature		temperature within the canopy / soil (threshold TGMIN)	codetemprac
root density		standard profile / true density	coderacine
N effect on root distribution (coderacine = true density)			codazorac
trophic-linked production (coderacine = true density)		continuous link / threshold /no	codtrophrac
frost			
plantlet or emergence frost	y/n		codgellev
leaf frost at juvenile phase (till AMF)	y/n		codgeljuv
leaf frost at adult phase	y/n		codgelveg
flower/fruit frost (from FLO)	y/n		codgelflo
water			
water requirements		crop coefficient / resistance approach	codebeso
interception of water by foliage	y/n		codeintercept
Nitrogen			
calculation nitrogen requirements		dense canopies (initial) / isolated plants (new calculation)	codeplisoleN
calculation Nitrogen stress index (INN)		INN cumulated / INN instantaneous	codeINN
leguminous	y/n		codelegume
maximal fixation capacity (codelegume = yes)		constant / growth fonction	codefixpot
nitrogen effect on fruit/grain number	y/n		codazofruit

TEC Options (1) (technical)

12 options level 1

ModuleName	flag	Formalism	Parameter Name
sowing			
planting structure (if radiative transfer)	y/n		codetradtec
rules to prescribe the sowing date	y/n		codedecisemis
Phenological stages			
forcing	y/n		codestade
irrigation			
automatic calculation of irrigations	y/n		codecalirrig
date of irrigation (codecalirrig=no)		in sum of upvt / in julian days	codedateappH2O
location of irrigation		above foliage / under foliage / in the soil	codlocirrig
Fertilization			
date of fertilisation		sum of upvt / julian days	codedateappN
splitting fertilisation		amounts in absolute value / amounts in % of the total value	codefracappN
location of mineral nitrogen inputs		at soil surface / below soil surface	codlocferti
harvest			
method of harvest		cutting /picking	codceuille
number of pickings (codcueille=picking)		one at the end / many during the cycle	nbceuille
harvest decision		physiological maturity / water content / sugar content / nitrogen content / oil content	codrecolte
minimum.or.maximum (codrecolte=water content)		minimum / maximul	codeaumin
Decision of harvest for associated crops		Maturity of the earliest / Maturity of the both (2 dates)	coderecolteassoc
rules of harvest/moisture status of the soil	y/n		codedecirecolte

TEC Options (2)

9 options level 1

ModuleName	flag	Formalism	Parameter Name
Special techniques			
cut crop	y/n		codefauche
Method.of.cutting (codefauche=yes)		automatic / calendar in days /calendar in degree days	codemodfauche
mulch		no/plastic mulch	codepaillage
topping	y/n		codrognage
topping calendar (codrognage=yes)		fixed date / autromatic calculation	codcalrogne
thinning	y/n		codeclaircie
leaf removal	y/n		codeffeuil
location of leaf removal (codeffeuil=yes)		bottom of the canopy/ top of the canopy	codhauteff
leaf.removal.calculation (codeffeuil=yes)		automatic calculation/ fixed date	codcaleffeuil
pruning	y/n		codetaille
trellis system	y/n		codepalissage
greenhouse crop	y/n		codabri
soil modification by techniques (compaction-fragmentation)			
activation fragmentation	y/n		codeDST
activation compaction sowing/harvest	y/n		codeDSTtass
nb of layers affected by the compaction		1 layer/2 /layers	codeDSTnbcouche

The Stics version

Thank you for your attention